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WINTER ROOST SITE CHARACTERISTICS, MORPHOMETRICS, AND DIET OF NORTHERN SAW-WHET OWLS (*AEGOLIUS ACADICUS*) IN OPEN-COUNTRY HABITATS OF WESTERN MONTANA

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ABSTRACT—We recorded the roost site characteristics, morphometrics, and diet of wintering Northern Saw-whet Owls (*Aegolius acadicus*) in open-country landscapes of western Montana from 1990 to 2021. Roosts had an average canopy cover of >70% and apparently provided dense overhead concealment. Average tree height was 4.8 m ($n = 24$), and average roost height was 1.6 m ($n = 43$). Roost sites were used by different owls over multiple years, but we found no evidence of winter roost site fidelity between years. Wintering owls ($n = 31$) averaged 11% heavier than the average mass of Saw-whet Owls we banded during autumn migration in western Montana ($n = 1830$). Pellet dissection ($n = 306$) yielded 481 prey items, consisting of 5 species, primarily *Microtus* voles (55%), and Deer Mice (38%). Paired long bones (for example: fibula-tibia, femur), were best for tallying numbers of prey, yielding >2 times as many prey items as skulls. Length × width for 2 sample groups of pellets was 30.5 × 14.9 ($n = 170$) and 31.9 × 14.6 ($n = 136$). Saw-whet Owl food niche breadth (H′ = 3.173 and D = 2.902) and dietary evenness (F = 0.8753) scores at the species level of resolution were relatively narrow, indicating that the owls consumed only a limited number of prey species of the total number of species that comprised their diet. Roost sites likely provided protection from avian predators, relief from winter weather, and avoidance of mobbing by small birds.

Key words: *Aegolius acadicus*, Montana, morphometrics, Northern Saw-whet Owl, prey selection, roost site selection, winter

Northern Saw-whet Owls (*Aegolius acadicus*), hereafter Saw-whet Owl, are common and widespread throughout much of North America (Rasmussen and others 2020). They are distributed from the southern coast of Alaska, east through Canada to Maine, south into North Carolina, and west to southern California. The Saw-whet Owl is known to be highly migratory, wintering in some areas in the southern portion of the range where breeding does not occur. In western Montana, Saw-whet Owls occur year-round, exhibiting breeding, migration, and wintering behaviors.

Saw-whet Owls are commonly associated with forested habitats containing tree cavities during the nesting season, while their wintering habitat is known to include dense woodlands, shrub-steppe, orchard, and riparian habitats (Hayward and Garton 1984; Grove 1985; Swengel and Swengel 1987, 1992a; Dinsmore and Clark 1991; Churchill and others 2000, 2002). In Montana, Saw-whet Owls are primarily associated with Black Cottonwood (*Populus trichocarpa*) bottomlands along rivers, mixed coniferous forests, and Aspen (*Populus tremuloides*) stands (Holt and Hillis 1987; Holt and Leroux 1996; Marks and others 2016). Despite numerous winter season studies (Hayward and Garton 1984, 1988; Grove 1985; Dinsmore and Clark 1991; Churchill and others 2000, 2002; Rasmussen and others 2020), we found no studies documenting riparian woody draws within grasslands as wintering habitat. Woody draws within open landscapes such as grasslands are relatively common and provide natural cover for many wildlife species. Yet, little is known regarding the use of woody draws within grasslands by Saw-whet Owls.

Nonetheless, woody draws, also known as riparian draws, have been declining in the western United States, largely due to livestock grazing, crop agriculture (Obe ki and others AQ: 2
2001; D'antonio and others 20 and housing and business development. In fact, <2% of all riparian habitat in the United States remains

natural and unaltered (Green and Kauffman 198[illegible]f the numerous published studies on Saw-whet Owl diet (Randle and Austing 1952; Catling 1972; Boula 1982; Smith and Devine 1982; Grove 1985; Cannings 1987; Swengel and Swengel 1987, 1992b; Hayward and Garton 1988; Marks and Doremus 1988; Dinsmore and Clark 1991; Hobson and Sealy 1991; Holt and others 1991; Holt and Leroux 1996; Whalen and others 2000; Underwood and Sealy 2002; Engel and others 2015; Rasmussen and others 2020), only 3 have documented winter-specific diet in western North America (Grove 1985; Holt and Leroux 1996; Woodruff and others 2020). While diet varies seasonally and with habitat type, it is inconclusive whether this species chooses their prey based on abundance at their breeding and wintering locations, or whether the location is chosen based on preferred prey availability (Hayward and Garton 1988; Johnson and Anderson 2003).

The main objective [illegible]was to examine roost site characteristics of Saw-whet Owls' wintering in deciduous woody draws within grasslands in western Montana. Additionally, we recorded owl morphometrics and feeding ecology.

Methods

Study Area

Our study is located in Missoula (UTM: Zone 11T, 718409.433E, 5200036.665N, WGS-84) and Mission valleys (UTM: Zone 11T, 712837.726E, 5278436.012N, WGS-84) of western Montana. The valleys are characterized by farm, ranch, and conservation lands interspersed with marshes, lakes, ponds, rivers, creeks, and irrigation reservoirs. Owl roosts were generally confined to woody draws with creek bottoms, dense patchy thickets, and planted shelterbelts within rangelands made up of grassland, shrubland, and cropland. Vegetation is a mixture of native and introduced trees, grasses, and shrubs.

Locating Owls

Our study area is entirely within the long-term breeding and non-breeding season monitoring of Long-eared Owl (*Asio otus*) populations, now in its 37th year (1986 to 2023). While collecting data on Long-eared Owls, Saw-whet Owls were incidentally encountered during winter surveys and Christmas bird counts. Consequently, we actively searched for Saw-whet Owl roosts every winter from 2011–2021. To locate roosting owls, we transected linear woody draws with anywhere from 2 to 4 people. Locating owls was easier and more successful in the absence of snow due to the visibility of fecal waste, or "whitewash." Nonetheless, we inspected almost all trees in each transect, visually scanning for fecal waste and body contour feathers on the ground, tree trunks, and branches, and over time developed a search image of the tree architecture Saw-whet Owls were most often found roosting in.

Stands of Ponderosa Pine (*Pinus ponderosa*), Douglas-fir (*Pseudotsuga menziesii*), and isolated Rocky Mountain Juniper (*Juniperus scopulorum*) were present, but our study area is primarily a grassland community dissected by deciduous woody draws. We did not search the tall coniferous trees for roosting owls, and thus possibly missed some owls.

We regularly found Great Horned Owls (*Bubo virginianus*), Saw-whet Owls, and Long-eared Owls. Less commonly we found Barn Owls (*Tyto alba*), Barred Owls (*Strix varia*), and only once, a Boreal Owl (*Aegolius funereus*). When a roost was located but no owl was present, species was determined through feathers, whitewash, and pellets. Saw-whet Owl pellets were easy to differentiate from the larger owl species by size. All the owl species reported herein produce buffy-colored whitewash, which was a key indicator of owl roost sites.

We often located individual and communal roosts of Black-billed Magpie (*Pica hudsonia*). To differentiate between owl and Magpie roosts we noted color differences in whitewash. Magpie whitewash was chalky white with a light purple tinge, likely from berries of hawthorn (*Crataegus* spp.) and Chokecherry (*Prunus virginianus*) trees that Magpies ate. Furthermore, Magpies usually roosted communally. Thus, whitewash was spread over a large area of ground and branches. Magpies were rarely roosting during the day, while Saw-whet Owls always roosted during the day and always alone. To differentiate Magpie pellets from Saw-whet Owl pellets, we noted that Saw-whet Owl pellets were made-up of fur and bones and had a spongier texture, whereas Magpie pellets often contained seeds and grasses, and were hard and slightly smaller (Owl Research Institute, Charlo, MT, unpubl. data; Reebs and Boag 1987). As Magpies also sometimes ate small mammals, determining roosting species was not

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FIGURE 1. Northern Saw-whet Owl winter roost in a hawthorn (*Crataegus* spp.) shrub in western Montana.

always reliable by pellets alone, in which case multiple forms of evidence were used.

Measuring Roost Sites

We measured winter roost sites from 2011 to 2021. We waited until the owls left the area during spring migration to measure roosts, but before spring vegetation growth started. Exact roost locations were noted through photographs (Fig. 1), repeated visits to the roost for trapping attempts, and the presence of feathers and whitewash on and below the roost branch; additionally, at times the bark or stem at the perch location was visibly worn from repeated use. We did not measure roosts with minimal evidence of owls as we believe they were only used briefly for a day or We considered a dedicated roost when there were piles of pellets and lots of whitewash. Sometimes multiple roosts were found within the same woody draw, but no more than 1 Saw-whet Owl was ever found occupying the same roost site.

Roost site characteristics were recorded as follows: tree species or genera; tree height; tree diameter at breast height (DBH); roost branch diameter at the perch location; roost height; height to canopy; canopy cover; and distance to open habitat. Not all measurements were collected from every roost due to human error, equipment availability, feasibility with vegetative structure, or some measurements only being collected in later years.

We used a spherical densiometer to measure the canopy cover (vanity) above the exact perch location using a ladder or by climbing the tree to give a truer measure of canopy concealment above the roosting owl, versus traditional methods of measuring general tree canopy cover from locations on the ground under the tree foliage. We used a meter tape to record the height of the roost from the highest point on the ground from which the tree trunk emerged. Height of the tree was determined with a clinometer when possible. Height to canopy represented the empty space between the roost branch and the cover (branches, twigs) above and was measured with a cm ruler. Distance to open habitat from the roost tree was estimated in meters. Most roost site coordinates were recorded with GPS.

Random sampling for roost locations was not recorded. Because roost sites were measured at the precise location of the owl on the branch, random sampling of specific branches proved difficult to implement.

Winter Body Measurements

We opportunistically trapped Saw-whet Owls using mist nets or by hand. Body mass, wing chord, tail length, molt, and relative age were recorded. We were not able to determine the sex of owls because there is currently no reliable method for sexing using morphometrics or plumage characteristics in this region. As a sexually dimorphic species, this creates a wide range in measurements. All owls were banded by authorized banders with number 4 butt-end, USGS aluminum leg bands. Recapture attempts were made within and between years at known roost sites.

Pellet Analysis

We collected as many Saw-whet Owl pellets as we could find for determination of food habits. We dissected pellets and pellet fragments to separate bones of small mammalian prey the owls ate. Prey species were determined from skulls and jaws, as described by Hoffman and Pattie (1968) and Foresman (2012).

We also separated and counted the large, paired bones from pellets. These included jaws, humeri, ulnas, radii, pelvises, femurs, scapulas, fibulae, and tibiae. We counted the fibula-tibia as 1 bone because they are fused, though the thin fibula is often broken off. Although the radius and ulna are often attached by tissue or membranes, they usually are found separated in pellets. Therefore, we counted them individually. We determined prey quantity by bone

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type by dividing paired bones by 2 to get a value of 1 prey item. For example, 2 jaws or femurs equals 1 prey item. We also measured length × width dimensions for 2 groups of pellets. We compared the number of large, paired bones with skulls to determine what method is best to identify and quantify prey species and numbers. However, we only used skulls to identify prey species.

We reported frequency and percentage of occurrence of species eaten. In order to analyze proportions, we selected the G-test, with significance levels set at 0.05. To meet assumptions of the test, we eliminated any category (for example: species, genera or group) that tallied <5 individuals (Zar 2019).

Food Niche Breadth and Dietary Evenness

Food niche breadth (FNB) and dietary evenness (DIEV) were determined using diversity indices. We chose Shannon-Weaver and Simpson's equations to examine species richness. There are strengths and weaknesses to each equation (see Zar 2019; Marti and others 2007; Collier and Schwertner 2012). Thus, we chose both equations because researchers often choose 1 equation over another. Overall, these equations allow large data sets to be expressed as single values. In turn, these values can be used for comparisons with similar studies, provided similar methods are used. FNB using Shannon-Weaver was calculated as $H' = -\Sigma p_i \log p_i$, where p_i represents the proportion of each species in the sample. The antilog of Shannon-Weaver's diversity index was used because it is linearly related to the number of prey categories in the sample. FNB using Simpson's equation was calculated as $D = \Sigma p_i^2$, where p^i is the proportion of each member of the assemblage being investigated. We used the reciprocal (1/D) of Simpson's index, because it's believed to yield a more meaningful number. DIEV scores were calculated using Hill's equation, $F = (N_2 - 1)/(N_1 - 1)$, where N_1 is the antilog of the Shannon-Weaver index (H') and N_2 is the reciprocal of Simpson's index.

We defined feeding niche as the relationship between owls and their prey and followed Holt (1993), Cromrich and others (2002), and Detienne and others (2008) in describing FNB and DIEV. Because FNB and DIEV are influenced by levels of resolution, we attempted to identify all prey to species. However, that was not always easy, as stated above. Thus, we present 2 levels of resolution: all prey categories ($n = 7$), and only species category ($n = 5$).

TABLE 1. Tree species used by Northern Saw-whet Owls as winter roosts, 2011 to 2021.

Tree species	*n*	%
Crataegus spp.	35	81.4
Salix spp.	4	9.3
Juniperus spp.	2	4.6
Prunus virginiana	2	4.6
Total	43	

We defined a broad FNB as a high number of prey species, relatively equally distributed in the owls' diet (heterogeneous); and a narrow FNB as a low number of prey species, relatively unequally distributed in the owls' diet (homogeneous) (Holt and Leroux 1996; Detienne and others 2008). Because DIEV scores range from 0 to 1, we considered a score of 1 or approximating 1 a uniform representation of prey proportions in the diet, and scores <0.50 to represent non-uniform or shared distribution (i.e., multiple predator species relying on the same prey resources within a given habitat) of prey in the diet (Detienne and others 2008).

Results

Roost Characteristics

We located and measured 43 individual roost sites from 2011 to 2021. Roost substrate consisted of 4 tree species: hawthorn ($n = 35$), willow (*Salix* spp.) ($n = 4$), juniper ($n = 2$), and chokecherry ($n = 2$). Hawthorns accounted for 81% of all roost sites (Table 1). T1

Roost sites had an average canopy cover percentage of $\bar{x} = 70.3$, $s = 14.6$, $n = 41$ at the exact point of the roost (Fig. 1). Roost sites averaged 1.6 m high ($n = 43$, $s = 6.7$), and roost trees averaged 4.8 m high ($n = 24$, $s = 2$). Roost height was on average 43.6% of the tree height. Diameter of the roost tree at breast height (DBH) averaged 270.3 mm ($n = 5$, $s = 85.4$), and roost branch diameter at the perching point had a wide range (9–40 mm) and averaged 20.5 mm ($n = 38$, $s = 11.9$). The height from the roost branch to the canopy above averaged 23.6 cm ($n = 33$, $s = 10$). Distance from the roost location to open habitat averaged 8.4 m ($n = 32$, $s = 4.2$) (Table 2). T2

TABLE 2. Northern Saw-whet Owl roost site characteristics in woody draws of grassland habitats of western Montana, 2011 to 2021.

Characteristic	*n*	Mean	±SD	SE	Range
Tree height (m)	24	4.8	2	0.41	2–10
Roost height (m)	43	1.6	6.7	1.02	0.7–5
Roost branch diameter (mm)	38	20.5	11.9	1.93	9–40
DBH (mm)	5	270.3	85.4	38.17	182.88–398.78
Height to canopy (cm)	33	23.6	10	1.74	10–56
Distance to open habitat (m)	32	8.4	4.2	0.74	2–20
% Canopy cover	41	70.3	14.6	0.02	34.32–91.52

Morphometrics and Site Fidelity

We trapped and banded 31 Saw-whet Owls during the wintering months (November to February). Average wing chord was 133.4 mm ($s = 4.4$, range = 124–144 mm), and average tail length was 72.7 mm ($s = 2.4$, range = 68–79 mm). Body mass averaged 95.4 g ($s = 13.4$, range = 74–135 mm) (Table 3).

Recapture attempts of banded Saw-whet Owls in the vicinity of their original roost location revealed at least 6 out of the 31 owls reused the exact same roost for months at a time within the same winter. Furthermore, surveys revealed different un-banded owls would use the same roost in subsequent years. We have no record of winter roost site fidelity for any individual owl in subsequent years.

Diet

We recorded 481 prey items from 306 pellets from at least 8 Saw-whet Owls. Five species of mammals represented 100% of the total prey (Table 4). *Microtus* voles accounted for 54.7% of total prey (Montane Voles [*Microtus montanus*, 28.9%]; Meadow Voles [*Microtus pennsylvanicus*, 18.3%]; and unknown *Microtus* vole [7.5%]). Deer Mice (*Peromyscus maniculatus*) made up 38.5% of total prey, shrews (*Sorex* spp.) 1.7%, and Northern Pocket Gophers (*Thomomys talpoides*) 0.4%. Some prey items could not be identified, with 4.8% identified as an unknown small mammal. Prey species proportions were significantly different ($G = 330.9$, $df = 5$, $P < 0.001$).

TABLE 3. Morphometric data of Northern Saw-whet Owls banded in winter (14 November 2014 to 15 February 2015), western Montana.

Measurement	*n*	Mean	± SD	Range
Wing chord (mm)	31	133.4	4.4	124–144
Tail length (mm)	31	72.7	2.4	68–79
Body mass (g)	31	95.4	13.4	74–135

Food Niche Breadth and Dietary Evenness

Using group level of identification ($n = 481$) for 7 categories, FNB using Shannon-Weaver index yielded H′ = 4.337, and Simpson's index D = 3.661. Hill's DIEV yielded F = 0.7975. When using species level of identification ($n = 422$) for 5 categories, FNB using Shannon-Weaver index yielded H′ = 3.173, and Simpson's index D = 2.902. Hill's DIEV yielded F = 0.8753.

Prey per Pellet

Bone frequencies were recorded for a sample of 220 pellets from at least 8 owls between 1990 and 1992. Fibula-tibia and femurs occurred in pellets most frequently (15.8 and 15.4%, respectively), followed by humeri, ulna, pelvis and radius. Jaws, skulls, and scapula were the least numerous in pellets, accounting for a combined 14.1% frequency, though it is worth noting that scapula was only recorded for 170 pellets from 5 roosts (Table 5).

Aside from skulls, other bones are naturally present in prey in pairs; therefore, prey quantity by bone type represents a truer measure of prey numbers from pellets than skulls alone. The typical method of estimating prey quantity using skulls resulted in 69 individuals, whereas estimating prey quantity using fibula-tibia resulted in 156

TABLE 4. Prey of Northern Saw-whet Owls removed from 306 pellets collected between November to February 1990–1992.

Species	*n*	% of total
Deer Mouse (*Peromyscus maniculatus*)	185	38.5
Montane Vole (*Microtus montanus*)	139	28.9
Meadow Vole (*Microtus pennsylvanicus*)	88	18.3
Unidentified *Microtus* spp.	36	7.5
Unidentified small mammal spp.	23	4.8
Shrew (*Sorex* spp.)	8	1.6
Northern Pocket Gopher (*Thomomys talpoides*)	2	0.4
Total	481	

TABLE 5. Bone contents from 220 complete pellets from at least 8 Northern Saw-whet Owls, November to February 1990–1992. PPP represents prey per pellet.

Bones	n	% Frequency	Prey quantity by bone type	PPP by bone type
Skull	69	3.5	69	0.31
Jaw	143	7.2	72	0.33
Fibula-Tibia	312	15.8	156	0.71
Radius	236	11.9	118	0.54
Ulna	285	14.4	143	0.65
Femur	305	15.4	153	0.70
Pelvis	270	13.7	135	0.61
Humerus	288	14.6	144	0.65
Scapula*	67	3.4	34	0.20
Total	1975			

*Scapula was only recorded for 170 pellets from 5 owls; prey per pellet by bone type reflects the correct number of pellets.

individuals, a 126% increase. Indeed, prey quantity represented by various bone counts differed drastically depending on which bone was used. In fact, there was an increase in estimated prey quantity of 64% to 271% when counting fibula-tibia ($n = 156$), femur ($n = 153$), humerus ($n = 144$), ulna ($n = 143$), pelvis ($n = 135$), or radius ($n = 118$) compared to when counting jaw ($n = 72$), skull ($n = 69$), or scapula ($n = 42$, adjusted). The difference in prey quantity by bone type was significant ($G = 143.0$, $df = 8$, $P < 0.001$).

Similar to prey quantity by bone type, prey per pellet (PPP) varied significantly depending on the type of bone observed. While the traditional method of counting skulls in pellets resulted in a PPP value of 0.23, pellet analysis using fibula-tibia and femurs represented PPP values of 0.71 and 0.70, respectively. In fact, there was a PPP range of 0.20–0.71 depending on bone type being counted (Table 5).

Pellet Length × Width

Mean pellet dimensions for a sample of 170 pellets were length = 30.5 mm ($s = 5.9$) and width = 14.9 mm ($s = 1.7$). Range was not recorded. Mean pellet dimensions for a sample of 136 pellets were length = 31.9 mm ($s = 7.47$, range = 19.2–50.2 mm), and width = 14.62 mm ($s = 2.3$, range = 10.4–20.8 mm).

Discussion

Roost Site

Saw-whet Owls appeared to select roost sites concealed within the tree architecture with an average canopy cover of over 70%, and roost heights of ≤5 m ($\bar{x} = 1.6$ m, range = 0.7–5 m). Our method of identifying and measuring roost branch diameter and canopy cover just centimeters above the owls' head gives a more detailed assessment of microhabitat selection, rather than measuring the entire tree canopy/foliage from selected points on the ground. Nonetheless, our findings are consistent with findings in other habitat types. For example, Swengel and Swengel (1992a) concluded Saw-whet Owls ($n = 623$ roosts) in coniferous forests of southern Wisconsin prefer roosts with high concealment (dense foliage concealing the view from ¾ of all angles), and at heights of ≤5 m. Additionally, results from measuring Saw-whet Owl roost sites in Idaho ($n = 15$, Hayward and Garton 1984) and Wisconsin ($n = 623$, Swengel and Swengel 1992a) showed that roost height positively correlated with tree height, and roost sites were higher in trees that provided less foliage cover on lower branches.

A recent study over 2 winter seasons in northwest Arkansas (Pruitt and others 2024) used radio telemetry to locate 13 individual roost sites for 7 Saw-whet Owls. All roost sites were high in coniferous trees, averaging about 18 m. The authors believed these roosts offered thermal advantages for warming and cooling, as well as potential protection from predation.

In a geographically related study from western Montana, about 30-km south of our study area, Stone and Ramsey (2025) used radio telemetry to track migrating Saw-whet Owls over 2 seasons. In contrast to winter-focused studies, Stone and Ramsey (2025) focused on Saw-whet Owl migration that ended in late October. One must consider that migrating owls may choose roost sites differently, stopping and resting as needed until reaching and settling in a wintering area. Nonetheless, Stone and Ramsey (2025) recorded 56 roosts of which 41 (73%) occurred high in tall Ponderosa Pines. They visually located 20 owls to estimate roost height.

Stone and Ramsey (2025) suggested traditional methods to locate Saw-whet Owls, such as "passive" winter ground searches (Grove 1985; Swengel and Swengel 1992a; this study), and many others (see Rasmussen and others 2020) are biased, implying these field methods may not detect owls roosting high in trees. While this statement has merit, it does not detract

from the fact that some of these studies, including ours, conducted rigorous systematic search efforts and yielded visual confirmation of numerous winter roosting owls. In fact, Swengel and Swengel (1992a) recorded 623 roosts from at least 25 owls. That is a robust sample and likely reflects typical roost site characteristics in that habitat type.

Additionally, biases exist in most wildlife studies, as study sites or individual animals (i.e., owls) are not randomly selected from the population of interest. One could equally argue the owls tracked by Stone and Ramsey (2025) are also a biased sub-sample, as: (1) individual owls were lured to capture sites by auditory methods, and thus could be biased by age, sex, and fitness; (2) the owls outfitted with transmitters were not randomly selected, and may not represent the true population; and (3) the authors ended their study in October, thus biasing their results to migrants. Migrants may have different habitat and roosting requirements from wintering owls, which is a common behavior in many species of birds. There are strengths and weakness to all methods, yet each method provides valuable natural history data.

Site Fidelity and Morphometrics

No individual owls were confirmed to reuse the same roost location for multiple winters. In fact, the presence of un-banded owls in the same roost locations as banded owls in years prior suggests individual Saw-Whet Owls share the same roost characteristic requirements. For example, once the owls select a habitat suitable for a wintering site, they then move about the habitat and select the micro-site suitable for daytime roosting. The lack of individuals reusing the same roost location for multiple years in our study also suggests low winter-roost site fidelity. This is similar to other studies that find low breeding site fidelity and natal philopatry in Saw-whet Owls (Cannings 1987; Marks and Doremus 2000; Marks and others 2015), and some have suggested Saw-whet Owls may be nomadic (Bowman and others 2010; Beckett and Proudfoot 2011; Marks and others 2015). AQ: 1 However, recapture sample sizes in our study ($n = 31$) may currently be too small to make conclusions on site fidelity.

Wintering body mass averaged 95.4 g, which is 11% heavier than the average mass of 85.9 g recorded in western Montana during autumn migration ($n = 1830$, Owl Research Institute, Charlo, MT, unpubl. data). Although winter sample sizes are relatively small ($n = 31$), the increased mass in winter captures versus autumn migration data suggests some owls gain weight once on their wintering grounds. Additionally, based upon fat scores, Stone and Ramsey (2025) suggested that the owls they captured during autumn migration were in poor body condition. Results from both studies suggest migrants could be under average weight when they depart on migration or migration is energetically costly and uses body fat. Once on their wintering grounds, however, some individuals may increase body mass and accumulate fat for spring migration, while others may not.

Diet

Pellet dissection and analysis is a common and widespread method in determining owl diet and food habits. As most studies show, skulls are typically used to determine prey species and quantity. Theoretically, the same quantity of prey should be represented by each bone type found within pellets for species that eat whole prey. However, in our study PPP yielded drastically different results depending on the type of bone being counted. In fact, some bones (e.g., fibula, tibia, ulna, femur, and humerus) indicated prey quantities of over twice that when counting skulls. This would indicate analysis of pellet contents based on skulls alone may not be reliable in determining overall prey species, prey numbers, and PPP. Thus, one should use caution when quantifying prey based on skulls alone and consider both methods in determining small owl species diets based on pellet analysis.

Food Niche Breadth and Dietary Evenness

While we reported measurements of FNB and DIEV at 2 levels of resolution, values calculated using prey identified to species may represent more meaningful measurements. Our FNB values using prey species ($H' = 3.173$; $D = 2.902$) indicate a narrow trophic structure, with a small number of prey species unequally distributed in the owl's diet. These results are similar to other non-breeding season studies of Saw-whet Owls across North America (Randle and Austing 1952; Catling 1972; Smith and Devine 1982; Grove 1985; Holt and others 1991),

including Holt and Leroux (1996), who report a FNB value of H′ = 3.3 (n = 366) between 1987 and 1992 in west-central Montana. Our DIEV value (F = 0.8753) is similar, yet higher than all comparable non-breeding season diet studies except for Holt and Leroux (1996), who reported a DIEV value of 0.89. Other studies in Ohio, Ontario, Connecticut, and Washington report DIEV values ranging from 0.58 to 0.742 (Randle and Austing 1952; Catling 1972; Smith and Devine 1982; Grove 1985; Holt and others 1991). These values indicate a small number of prey items were eaten in similar proportions across all study sites.

Summary

Why did Saw-whet Owls in our study select low roost sites with dense overhead cover, and what appeared to be one-way in, one-way out configuration? We suggest several hypotheses, neither mutually exclusive. First, we believe predation is the primary driver of winter roost site selection. The umbrella-like cover and surrounding branches provided by roost sites could serve as protection from avian predators such as other larger owl species, accipiter hawks, Magpies, or perhaps some mammalian predators, which all occur in the area. In fact, we have documented Great Horned Owl and Barred Owl predation on Saw-whet Owls during our studies. Second, selection for these deciduous woody draws may have thermal benefits as the lower elevation creek bottoms have steep banks below the relatively flat grasslands. By roosting below the grasslands plain, the owls may gain protection or relief from cold winter temperatures, and strong unobstructed winter winds blowing across the grasslands. Third, the draws also receive less snow cover, and thus may provide patchy open habitat for hunting small mammals. Finally, these roost sites may also provide relief from mobbing by mixed species flocks of small passerines that winter and forage within the draws. Collectively, these roost sites likely provide the owls with a safe daytime resting period.

Our study identifies woody draws within open-country grasslands as overwintering habitat for Saw-whet Owls in western Montana. In North America, grassland birds are declining at a steeper rate than any other group of birds. This is largely attributed to the conversion of grassland to farmland and grazing, resulting in habitat loss, degradation, and fragmentation (Sauer and others 2017). Over 80% of North America's grasslands have been developed or converted for agricultural purposes, and much of the remaining habitat is utilized for grazing (White and others 2000). Therefore, identifying habitat requirements for wildlife is crucial for developing conservation strategies and targeting areas for protection and management.

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